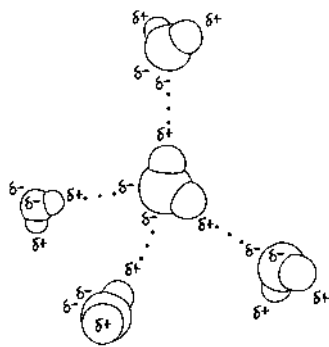


Figure 2.14 One possible answer:



Concept Check 2.1

1. Table salt (sodium chloride) is made up of sodium and chlorine. We are able to eat the compound, showing that it has different properties from those of a metal (sodium) and a poisonous gas (chlorine). 2. Yes. An organism requires trace elements, though only in small amounts. 3. A person with an iron deficiency will probably show fatigue and other effects of a low oxygen level in the blood. (The condition is called anemia and can also result from too few red blood cells or abnormal hemoglobin.) 4. Variant ancestral plants that could tolerate elevated levels of the elements in serpentine soils could grow and reproduce there. (Plants that were well adapted to nonserpentine soils would not be expected to survive in serpentine areas.) The offspring of the variants would also vary, with those most capable of thriving under serpentine conditions growing best and reproducing most. Over many generations, this probably led to the serpentine-adapted species we see today.

Concept Check 2.2

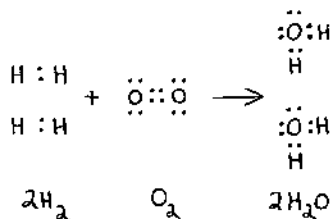
1. 7 2. ^{15}N 3. 9 electrons; two electron shells; 1s, 2s, 2p (three orbitals); 1 electron is needed to fill the valence shell. 4. The elements in a row all have the same number of electron shells. All the elements in a column have the same number of electrons in their valence shells.

Concept Check 2.3

1. In this structure, each carbon atom has only three covalent bonds instead of the required four. 2. The attraction between oppositely charged ions, forming ionic bonds. 3. If you could synthesize molecules that mimic these shapes, you might be able to treat diseases or conditions caused by the inability of affected individuals to synthesize such molecules—or to block the function of such molecules if overproduction is the cause of the disorder.

Concept Check 2.4

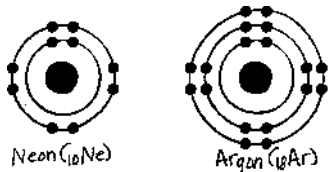
1.



2. At equilibrium, the forward and reverse reactions occur at the same rate. 3. $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$. Glucose and oxygen react to form carbon dioxide and water, releasing energy. We breathe in oxygen because we need it for this reaction to occur, and we breathe out carbon dioxide because it is a by-product of this reaction. (This reaction is called cellular respiration, and you will learn more about it in Chapter 9.)

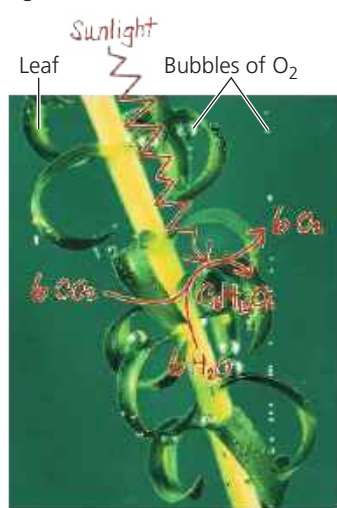
Summary of Key Concepts Questions

2.1 A compound is made up of two or more elements combined in a fixed ratio, while an element is a substance that cannot be broken down to other substances. 2.2



Both neon and argon have completed valence shells, containing 8 electrons. They do not have unpaired electrons that could participate in chemical bonds. 2.3 Electrons are shared equally between the two atoms in a nonpolar covalent bond. In a polar covalent bond, the electrons are drawn closer to the more electronegative atom. In the formation of ions, an electron is completely transferred from one atom to a much more electronegative atom. 2.4 The concentration of

Figure 2.17



products would increase as the added reactants were converted to products. Eventually, an equilibrium would again be reached in which the forward and reverse reactions were proceeding at the same rate and the relative concentrations of reactants and products returned to where they were before the addition of more reactants.

Test Your Understanding

1. D 2. A 3. B 4. A 5. D 6. B 7. C 8. D

9. a. $\begin{array}{c} \text{H} & \text{H} \\ | & | \\ \text{H} : \text{C} : \text{C} : \text{O} \\ | & | \\ \text{H} & \text{H} \end{array}$ This structure makes sense because all valence shells are complete, and all bonds have the correct number of electrons.
- b. $\begin{array}{c} \text{H} & \text{H} \\ | & | \\ \text{H} : \text{C} : \text{H} : \text{C} : \text{O} \\ | & | \\ \text{H} & \text{H} \end{array}$ This structure doesn't make sense because H has only 1 electron to share, so it cannot form bonds with 2 atoms.

Chapter 3

Figure Questions

Figure 3.2 One possible answer:

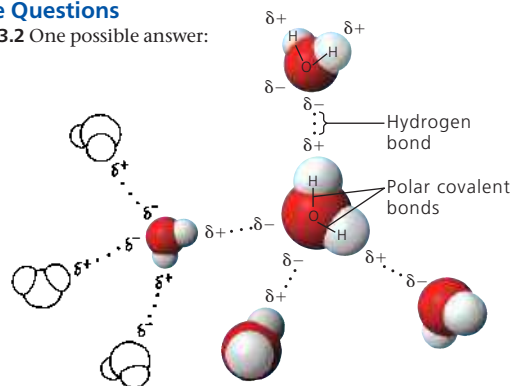


Figure 3.8 Heating the solution would cause the water to evaporate faster than it is evaporating at room temperature. At a certain point, there wouldn't be enough water molecules to dissolve the salt ions. The salt would start coming out of solution and re-forming crystals. Eventually, all the water would evaporate, leaving behind a pile of salt like the original pile. Figure 3.12 Adding excess CO_2 to the oceans ultimately reduces the rate at which calcification (by organisms) can occur.

Concept Check 3.1

1. Electronegativity is the attraction of an atom for the electrons of a covalent bond. Because oxygen is more electronegative than hydrogen, the oxygen atom in H_2O pulls electrons toward itself, resulting in two partial negative charges on the oxygen atom and a partial positive charge on each hydrogen atom. Atoms in neighboring water molecules with opposite partial charges are attracted to each other, forming a hydrogen bond. 2. Due to its two polar covalent bonds, a water molecule has regions of partial negative charge on the O atom that allow it to form hydrogen bonds with hydrogen atoms on neighboring water molecules, and regions of partial positive charge on the H atoms that allow them to form hydrogen bonds with oxygen atoms on neighboring water molecules. 3. The hydrogen atoms of one molecule, with their partial positive charges, would repel the hydrogen atoms of the adjacent molecule. 4. The covalent bonds of water molecules would not be polar, so no regions of the molecule would carry partial charges and water molecules would not form hydrogen bonds with each other.

Concept Check 3.2

1. Hydrogen bonds hold neighboring water molecules together. This cohesion helps chains of water molecules move upward against gravity in water-conducting cells as water evaporates from the leaves. Adhesion between water molecules and the walls of the water-conducting cells also helps counter gravity. 2. High humidity hampers cooling by suppressing the evaporation of sweat. 3. As water freezes, it expands because water molecules move farther apart in forming ice crystals. When there is water in a crevice of a boulder, expansion due to freezing may crack the boulder. 4. The hydrophobic substance repels water, perhaps helping to keep the ends of the legs from becoming coated with water and breaking through the surface. If the legs were coated with a hydrophilic substance, water would be drawn up them, possibly making it more difficult for the water strider to walk on water.

Concept Check 3.3

1. 10^5 , or 100,000 2. $[\text{H}^+] = 0.01 \text{ M} = 10^{-2} \text{ M}$, so $\text{pH} = 2$. 3. $\text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COO}^- + \text{H}^+$. CH_3COOH is the acid (the H^+ donor), and CH_3COO^- is the base (the H^+ acceptor). 4. The pH of the water should decrease from 7 to about 2 (as mentioned in the text); the pH of the acetic acid solution will decrease only a small amount, because as a weak acid, it acts (like carbonic acid) as a buffer. The reaction shown for question 3 will shift to the left, with CH_3COO^- accepting the influx of H^+ and becoming CH_3COOH molecules.